

Deep Learning

CSE641/ECE555

Lecture 2.1 | Take your own notes during lectures

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ML Recap

What do we actually mean by Learning?

Learning Process

Learning consists of finding a “good” function, that “does what is it supposed to do” (classifying, regressing, etc.)

- **Generator** of samples from a fixed but unknown distribution $P(x)$
- A **supervisor** (oracle) which returns an output, for every input sample, as per the conditional distribution $P(y|x)$, also fixed but unknown
- A **learning machine** capable of implementing a set of functions $f(x; \theta)$

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The Key here is **Parameterization!** Our so called **Mathematical model** consists of **parameters** that are initialized/adjusted/adapted-iteratively to achieve **Learning**

We say **Learning** is happening when a system (with our **model**) performs better at a **task** than before.

Learning via Optimizing Loss

Learning problem is to choose from the given set of functions the one which best approximates the supervisor's response.

$\mathcal{F}$ is the set of all the functions f that the learning phase may produce. For instance, when model is fixed, $\mathcal{F}$ contains one mapping per parameter configuration.

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How to choose the best function?

- **Randomly:**
 - Advantage: Really fast
 - Disadvantage: Terrible performance
- **Scan entire $\mathcal{F}$ and pick the best:**
 - Advantage: Great performance
 - Disadvantage: Terribly slow
- **Intelligent Way: Learn it using the performance metric**
 - Loss functions help to evaluate the performance to identify the optimal one.

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The **Loss** $\mathcal{L}$ is a function that indicates how bad a function f (e.g. classifier, regressor) is performing at its task. The loss may include additional terms related to f itself.

Loss functions are always non-negative.

Minimum loss means we are not making any mistakes.

Classification

$$\mathcal{L}(f, (x, y)) = \mathbf{1}_{f(x) \neq y}$$

Regression

$$\mathcal{L}(f, (x, y)) = (f(x) - y)^2$$

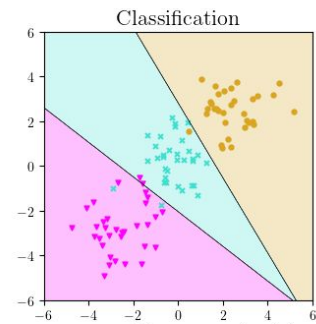
Density Estimation

$$\mathcal{L}(q, z) = -\log q(z)$$

Learning: Probabilistic Perspective (IID Samples)

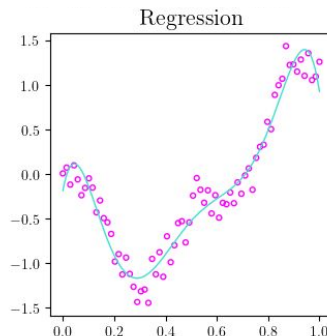
Classification,

- (X, Y) random variables on $\mathbb{R}^D \rightarrow [1 \dots C]$
- we want to estimate $y: \max P(Y = y|X = x)$
-



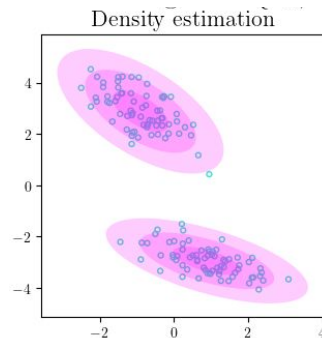
Regression,

- (X, Y) random variables on $\mathbb{R}^D \rightarrow \mathbb{R}$
- we want to estimate $\mathbb{E}(Y|X = x)$



Density estimation,

- X random variable on $\mathbb{R}^D$
- we want to estimate μ_X



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Regression

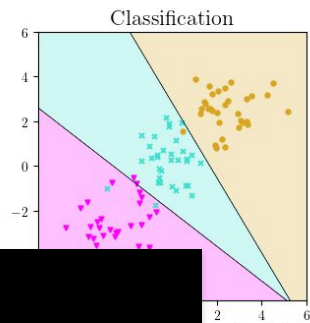
- (X, Y) r
- we want

The relationships are fuzzy:

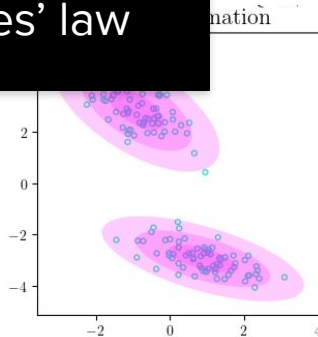
- Regression allows to do classification through class scores
- Density models allow to do classification thanks to Bayes' law

Density estimation,

- X random variable on $\mathbb{R}^D$
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Regression



Generalization: Expected and Empirical Risk

We are looking for an f with a small **unknown expected risk**

$$\mathbb{R}(f) = \mathbb{E}_{P_{X,Y}} [\mathcal{L}(f(X), Y)]$$

which requires optimizing

$$f^* = \operatorname{argmin}_{f \in \mathcal{F}} \mathbb{R}(f)$$

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For IID training samples we can approximate expected risk via **empirical risk**

$$\mathcal{D} = [(x_1, y_1) \dots (x_n, y_n)]$$

$$\hat{\mathbb{R}}(f; \mathcal{D}) = \hat{\mathbb{E}}_{\mathcal{D}}(\mathcal{L}(f(X), Y)) = \frac{1}{N} \sum_{n=1}^N \mathcal{L}(f(x_n), y_n)$$

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The **law of large numbers** (LLN) states that the **average** of the results obtained from a large number of **independent** random samples **converges** to the true value, **if it exists**.

For III

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The **expected risk** is the **expectation of the loss** when the data follows the true distribution of $\mathbf{Z}$, and in **practice we don't have access to the true distribution of $\mathbf{Z}$** .

How Well Can We Do? $y = f(x)$ $\mathbb{R}(f) = \mathbb{R}_N^{emp}(f) + \text{overfit penalty}$

- ★ We define the **capacity** of a model (set of predictors) as its ability to approximate an **arbitrary function**. This is a vague definition in general.
- ★ A mathematically precise notion is the **Vapnik–Chervonenkis dimension** of a set of functions, which, in the Binary classification case, is the cardinality of the largest set that can be labeled arbitrarily (Vapnik, 1995). **Learn more about this in Learning Theory Course.**
- ★ It is a very powerful concept, but is poorly adapted to high dimensional problems.

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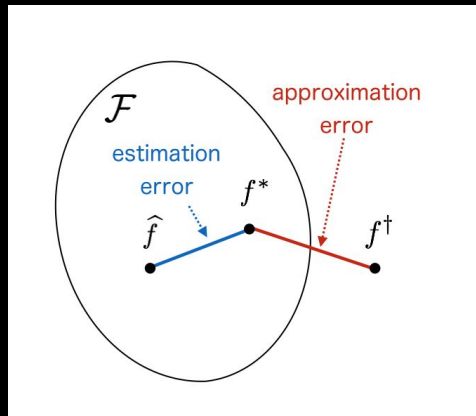
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- ★ **More Capacity $\neq$ Better Model**
- ★ **Controlling Over-fitting**
 - **Reducing the space** (less functionals, simple DNN with fewer modules/layers)
 - **Making the choice of f^* less dependent on data** (penalty on coefficients, margin maximization, ensemble methods)

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Convergence rate depends on the number of samples! **Optimal (parametric) convergence rate**

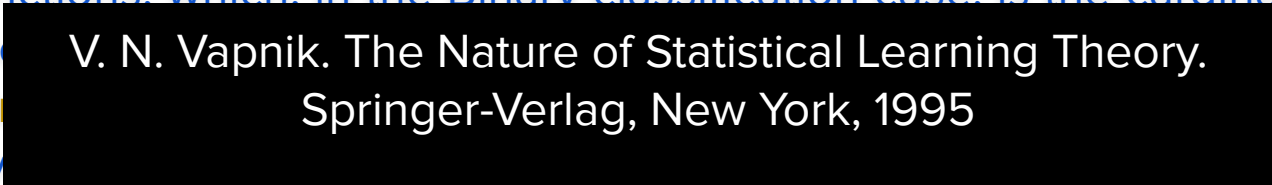
Can we do better? Not without additional assumptions. **More in coming lectures.**



$$\sup_{T \in \mathcal{F}} (\mathbb{E}_X \mathbb{E}(\mathcal{L}(\hat{f}, T) | x_1 \dots x_n) - \inf_{f \in \mathcal{F}} \mathbb{E} \mathcal{L}(f, T)) \geq \frac{c}{\sqrt{n}}$$

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V. N. Vapnik. The Nature of Statistical Learning Theory. about this
in Learning Springer-Verlag, New York, 1995
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- ★ A mathematical set of functions of the largest size in **Learn**

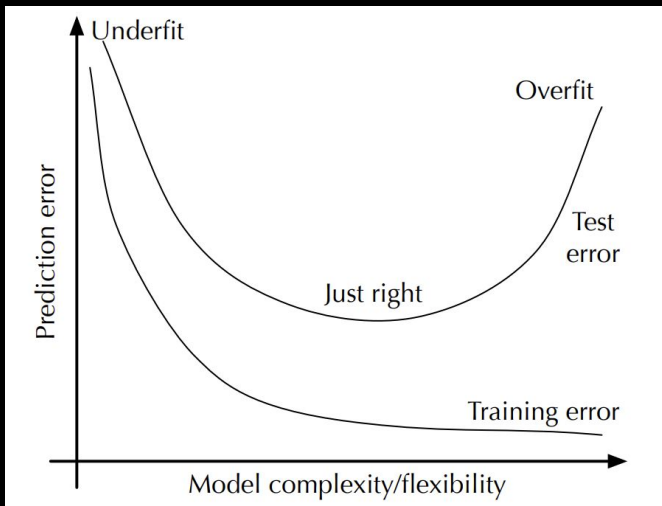
- ★ It is a very hard problem

- ★ **More Capacity**
- ★ **Control**

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Two important factors determining generalization ability:

- Model complexity
- Training data size



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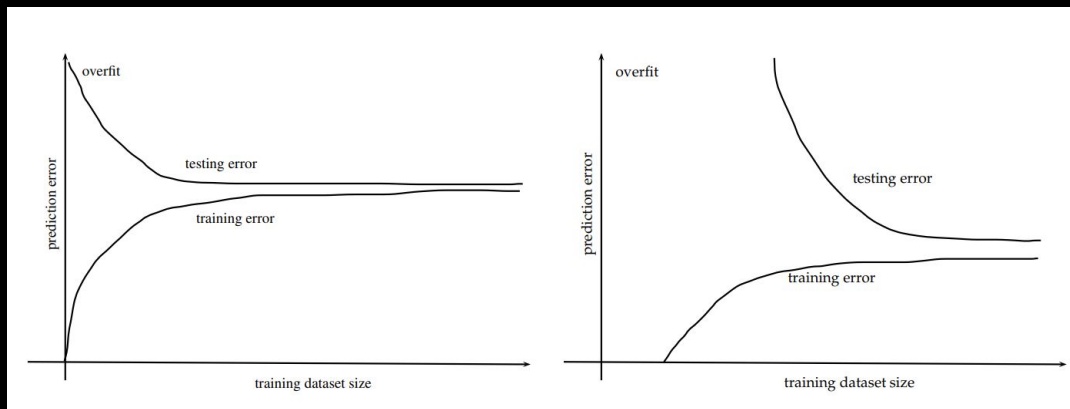
- ★ A mathematical model (set of functions) that can approximate the largest set of functions is called the **universal approximation** of a function.

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- ★ More Capacity
- ★ Controlling

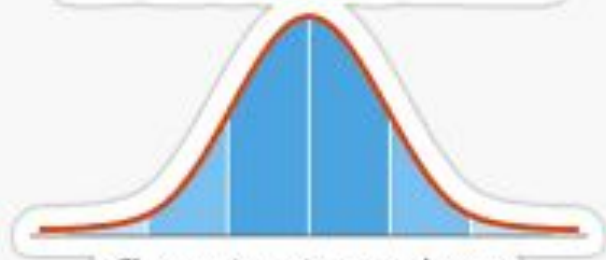
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The normal distribution



**The only place where
average is considered good**